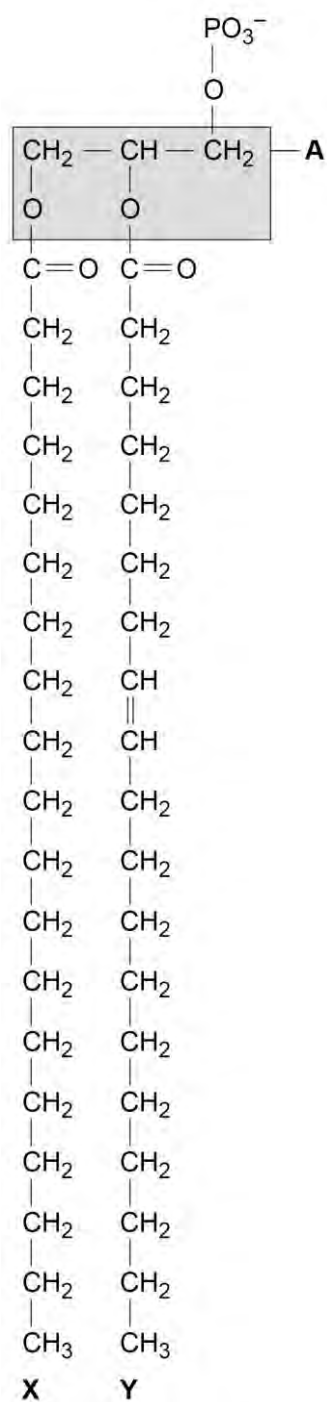


0 7 . 1 Describe how you would test a piece of food for the presence of lipid.

[2 marks]

Figure 8 shows a phospholipid.

Figure 8



- 0 7** . **2** The part of the phospholipid in **Figure 8** labelled **A** is formed from a particular molecule. Name this molecule.

[1 mark]

- 0 7** . **3** Name the type of bond between **A** and fatty acid **X**.

[1 mark]

- 0 7** . **4** Which of the fatty acids, **X** or **Y**, in **Figure 8** is unsaturated? Explain your answer.

[1 mark]

Question 7 continues on the next page

Scientists investigated the percentages of different types of lipid in plasma membranes from different types of cell. **Table 2** shows some of their results.

Table 2

Type of lipid	Percentage of lipid in plasma membrane by mass		
	Cell lining ileum of mammal	Red blood cell of mammal	The bacterium <i>Escherichia coli</i>
Cholesterol	17	23	0
Glycolipid	7	3	0
Phospholipid	54	60	70
Others	22	14	30

07 . 5

The scientists expressed their results as **Percentage of lipid in plasma membrane by mass**. Explain how they would find these values.

[2 marks]

Cholesterol increases the stability of plasma membranes. Cholesterol does this by making membranes less flexible.

07 . 6

Suggest **one** advantage of the different percentage of cholesterol in red blood cells compared with cells lining the ileum.

[1 mark]

0 7 . 7 *E. coli* has no cholesterol in its cell-surface membrane. Despite this, the cell maintains a constant shape. Explain why.

[2 marks]

Turn over for the next question